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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/579,529	08/16/2012	Taizo Uda	HARAK-32816/US-1/PCT	2407

72960 7590 07/01/2014

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ART UNIT	PAPER NUMBER
1648	

MAIL DATE	DELIVERY MODE
07/01/2014	PAPER

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Office Action Summary	Application No. 13/579,529	Applicant(s) UDA ET AL.	
	Examiner M. FRANCO SALVOZA	Art Unit 1648	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☒ Responsive to communication(s) filed on *3/20/2014*.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL**. 2b)☒ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5)☒ Claim(s) *1-7 and 25* is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6)☐ Claim(s) _____ is/are allowed.

7)☒ Claim(s) *1-7 and 25* is/are rejected.

8)☐ Claim(s) _____ is/are objected to.

9)☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☒ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date. _____ .

4) ☐ Other: _____.

The present application is being examined under the pre-AIA first to invent provisions.

DETAILED ACTION

1. Claim 4 has been amended. Claims 1-7, 25 are under consideration.

2. It is noted that applicant has filed three sets of claim amendments (3/17/2014; 3/19/2014; 3/20/2014). The most recent version filed 3/20/14 cumulatively incorporating all of the amendments is the claim set under consideration.

3. In light of the new rejections below, this Action is a Non Final Action.

Claim Objections

4. (**previous objection, withdrawn**) Claims 3, 4 were objected to because of informalities.

In light of applicant's amendments the objections is withdrawn.

Claim Rejections - 35 USC § 103

5. (**previous rejection, withdrawn**) Claims 1, 4 were rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Thorpe et al. in view of Lowman et al. and Smider et al. (previously cited).

Amended claim 4 recites a human abzyme comprising an unconjugated human antibody kappa (κ) light chain with a variable domain consisting of SEQ ID NO: 30, wherein the human abzyme has an anti rhabdovirus activity, and cytotoxicity against cancer cells.

Applicant contends that: claim 1 was amended to no longer recite SEQ ID NO: 30; applicant amended claim 4 and contends that amended claim 4 is not obvious by any combination of Thorpe, Lowman and/or Smider; the abzyme is different from the immunoconjugate disclosed in Thorpe, Lowman and/or Smider in that the abzyme is not conjugated with another preparation or antibody, and is effective alone; any of Thorpe, Lowman and/or Smider do not disclose or suggest the abzyme in claim 4 alone has such an effect; an antibody light chain containing A3-JK1 in Smider et al. has an endpeptidase activity and is used in combination with a heteroantibody heavy chain having a specificity with respect to a target protein; Smider et al. does not disclose or suggest that the antibody light chain alone has a specificity with respect to a rhabdovirus or a cancer.

In light of applicant's amendments the rejection is withdrawn.

Claim Rejections - 35 USC § 102

The following is a quotation of the appropriate paragraphs of pre-AIA 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

> (b) the invention was patented or described in a printed publication in this or a foreign country or in public use or on sale in this country, more than one year prior to the date of application for patent in the United States.

6. (**new rejection**) Claim 4 is rejected under pre-AIA 35 U.S.C. 102(b) as anticipated by Smider et al. (previously cited).

See the recitation to amended claim 4 above.

See the teachings of Smider et al. in previous Office Actions. To reiterate, Smider et al. teaches catalytic polypeptide for treating viral infections and cancers; including human antibody chains [0002]; including human proteolytic light chain A3-JK1 (the K is interpreted to mean "kappa"), which has 100% homology to instant SEQ ID NO: 30 (See Result 1 (ADO07297) in Search Results 2013-517_160905_us-13-579-529-30.rag in SCORE)). Thus the sequence is interpreted as a human antibody kappa light chain with a variable domain consisting of SEQ ID NO: 30.

Since the specification teaches wherein the antibody κ light chain is also referred to as abzyme (p. 17), the term "human abzyme" is interpreted to also mean merely the human antibody κ light chain.

Smider et al. teaches wherein the catalytic polypeptide treats viral infection and cancer. It is noted that Smider et al. does not teach wherein the human abzyme has an anti rhabdovirus activity, and cytotoxicity against cancer cells. However, since Smider et al. teaches the identical structure of the human antibody kappa light chain, said functional characteristics are presumed to be inherent. Where the claimed and prior art products are identical or substantially identical in structure or composition, or are produced by identical or substantially identical processes, a *prima facie* case of either anticipation or obviousness has been established. *In re Best,* 562 F.2d 1252, 1255, 195 USPQ 430, 433 (CCPA 1977). "When the PTO shows a sound basis for believing that the products of the applicant and the prior art are the same, the applicant has the burden of showing that they are not." *In re Spada,* 911 F.2d 705, 709, 15 USPQ2d 1655, 1658 (Fed. Cir. 1990)(See MPEP 2112.01).

Therefore, since Smider et al. alone teaches the human antibody kappa light chain used in catalytic antibody with the identical amino acid sequence of instant SEQ ID NO: 30, the teaching of Smider et al. is asserted to meet the claim limitations in light of the definition included in the specification.

Claim Rejections - 35 USC § 101

35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

7. (**new rejection**) Claims 1-7, 25 are rejected under 35 U.S.C. 101 because the claimed invention is directed to non-statutory subject matter because the claimed invention is not directed to patent eligible subject matter.

Based upon an analysis with respect to the claim as a whole, claim(s) 1-7, 25 do not recite something significantly different than a judicial exception. The rationale for this determination is explained below:

Claim 1 recites a human abzyme comprising a human antibody kappa (κ) light chain with a variable domain consisting of SEQ ID NO: 26, 14, 50, 35, 54, or 22.

Claim 2 recites the human abzyme as set forth in Claim 1, wherein: the human abzyme has an anti rhabdovirus activity, and anti influenza virus activity; and the variable domain consists of SEQ ID NO: 26.

Claim 3 recites the human abzyme as set forth in Claim 1, wherein: the human abzyme has an anti rhabdovirus activity, anti influenza virus activity, and cytotoxicity against cancer cells; and the variable domain consists of SEQ ID NO: 14.

See the recitation to claim 4 above.

Claim 5 recites the human abzyme as set forth in Claim 1, wherein: the human abzyme has an anti influenza virus activity and nucleolytic activity; and the variable domain consists of SEQ ID NO: 50.

Claim 6 recites the human abzyme as set forth in Claim 1, wherein: the human abzyme has an anti influenza virus activity; and the variable domain consists of SEQ ID NO: 35.

Claim 7 recites the human abzyme as set forth in Claim 1, wherein: the human abzyme has an anti influenza virus activity; and the variable domain consists of SEQ ID NO: 54.

Claim 25 recites the human abzyme as set forth in Claim 1, wherein: the human abzyme has an anti rhabdovirus activity; and the variable domain consists of SEQ ID NO: 22.

The specification teaches that the human antibody κ light chain indicates a light chain of immunoglobulin (immune globulin) originating from a human (p. 11)(or of human origin (p. 18)); wherein known human antibody constant domain can be added (p 13; and thus, also, not added); wherein the antibody κ light chain is also referred to as abzyme (p. 17); wherein the polypeptide is immunoglobulin of human origin that exhibits amidase activities (p. 18); there exist variants of natural proteins that do not significantly alter the structure of the function of the natural protein (p. 19). Thus, the interpretation at its most basic is that the human abzyme comprising a human antibody kappa (κ) light chain with a variable domain consisting of SEQ ID NO: (26, 14, 50, 35, 54 or 22; or 30) reads upon just a human antibody kappa (κ) light chain with

a variable domain consisting of SEQ ID NO: (26, 14, 50, 35, 54 or 22; or 30), which, according to the specification, is of human origin.

Thus, claims 1-7, 25 recite product claims (here, antibody chains) that appear to be natural product that is not markedly different in structure from naturally occurring product. Merely isolating variable domains (as recited in claims 1-7, 25) do not create a marked difference in structure between the chains and their naturally occurring counterparts. Removal or isolation of the claimed chains does not change their structure.

Thus the human abzymes as recited in claims 1-7, 25 read on naturally occurring human antibody kappa light chain.

(See "2014 Procedure For Subject Matter Eligibility Analysis Of Claims Reciting Or Involving Laws Of Natural Principles, Natural Phenomena, And/Or Natural Products" (March 4, 2014) at http://www.uspto.gov/patents/law/exam/examguide.jsp).

Conclusion

8. As indicated in the previous Office Action of 12/24/2013, SEQ ID NO:s: 26, 14, 50, 35, 54, 22 are free of the prior art.

9. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

A. Welschof et al. (see 892-Notice of References Cited) teaches human immunoglobulin light chain variable region (See Result 1, 20130517_160905_us-13-579-529-30.rpr (S58207; submitted to the EMBL Data Library, July 1995)) having 100% homology to instant SEQ ID NO: 30.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to M. FRANCO SALVOZA whose telephone number is (571)272-4468. The examiner can normally be reached on Monday to Thursday, 9 a.m. to 7:30 p.m.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Zachariah Lucas can be reached on 571-272-0905. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/M. Franco Salvoza/

Patent Examiner, Art Unit 1648

/Zachariah Lucas/

Supervisory Patent Examiner, Art Unit 1648